5.1.2 Out into space

	Learning outcomes	Additional guidance
(a)	*Describe and explain:*	
	(i) changes of gravitational and kinetic energy	
	(ii) motion in a uniform gravitational field	
	(iii) the gravitational field and potential of a point mass	modelling the mass of a spherical object as a point mass at its centre
	(iv) angular velocity in rad s^{-1}	
	(v) motion in a horizontal circle and in a circular gravitational orbit.	
(b)	*Make appropriate use of:*	
	(i) the terms: force, kinetic and potential energy, gravitational field, gravitational potential, equipotential surface	HSW2, 8
	by sketching and interpreting:	
	(ii) graphs showing gravitational potential as area under a graph of gravitational field versus distance, graphs showing changes in gravitational potential energy as area under a graph of gravitational force versus distance between two distance values	*M3.8, M3.12* HSW5
	(iii) graphs showing force as related to the tangent of a graph of gravitational potential energy versus distance, graphs showing field strength as related to the tangent of a graph of gravitational potential versus distance	*M3.6* HSW5
	(iv) diagrams of gravitational fields and the corresponding equipotential surfaces.	
(c)	*Make calculations and estimates involving:*	
	(i) uniform gravitational field, gravitational potential energy change = *mgh*	Learners will also be expected to recall this equation
	(ii) energy exchange, work done, $\Delta E = F\Delta s$; no work done when the force is perpendicular to the displacement, resulting in no work being done whilst moving along equipotentials	*M2.1*
	(iii) $a = v^2/r$, $F = mv^2/r = mr\omega^2$	
	(iv) the radial components: $F_{grav} = -\frac{GmM}{r^2}$, $g = \frac{F_{grav}}{m} = -\frac{GM}{r^2}$	
	(v) gravitational potential energy $E_{grav} = -\frac{GMm}{r}$	